

Question				Marking details		Marks Available					
						AO1	AO2	AO3	Total	Maths	Prac
2	(a)			Alpha radiation → electromagnetic wave Beta radiation → helium nuclei Gamma radiation → high energy electrons	3 correct 2 marks 1 or 2 correct 1 mark 0 correct 0 mark	2			2		
	(b)			Indicative content: Collecting data Measure the background count of the room for 2 minutes. Using the tongs hold a source close, e.g. 2-3 cm to the GM tube and record the count for the same time. Now place the piece of paper between the source and GM tube or increase the distance in suitable intervals and record the count for 2 minutes. Repeat this technique for the other 2 sources. Analysing data The alpha source will have a count rate that reduces to background count (or 0 if background subtracted) when paper absorber present or distance increased e.g. 30 cm. The beta source will have a count rate that doesn't change when paper is used as absorber or distance increased e.g. 30 cm. The alpha/beta source will have a count rate that decreased when the paper absorber is inserted or distance increased but is higher than the background count.		3	3		6		6